

NASA Data Center Infrastructure RFI Ready Response - 2026-07-11

- Opportunity: **80TECH26RFI0020**
- Response type: RFI / market research capability response
- Submission posture: **READY_FOR_HUMAN_REVIEW_AND_EMAIL_STAGING**
- External send without human: **false**
- Final submission without human: **false**
- Pricing included: **false**
- Claim boundary: no NASA deployment, NASA validation, FedRAMP authorization, realized savings, or production access is claimed.

Suggested Subject

Response to RFI 80TECH26RFI0020 - LumenCore Proof-to-Pilot Validation Layer for AI-Ready Infrastructure Modernization

Executive Summary

LumenCore proposes a proof-to-decision validation layer for data center modernization and AI-ready infrastructure planning. The system helps mission owners compare modernization paths using bounded source records, locked baselines, candidate workflow replay, resilience and operational-readiness metrics, and reviewer-facing evidence manifests.

The value is not another dashboard or unbounded AI claim. The value is a reproducible decision record: what source data was used, what baseline was accepted, what candidate workflow was tested, what metric changed, what failed, what uncertainty remains, and what is ready for a controlled pilot.

For NASA, this approach can support market research around hybrid infrastructure, workload placement, AI operations, resilience planning, energy and cost proxies, and vendor-claim validation before large-scale operational commitments are made.

Company Overview

LumenCore is an early-stage AI infrastructure validation company led by Robert Ashworth. The company focuses on proof-to-pilot instrumentation for AI and data systems: source provenance, baseline lock, candidate-vs-baseline replay, evidence manifests, and human-reviewed pilot gates.

Current work includes a measured-source proof stack, federal submission readiness artifacts, and reviewer-safe evidence packets for technical, government, and funding audiences. The platform is designed to separate measured proof from assumptions, simulations, future-work claims, and unsupported deployment claims.

Relevant Capabilities

Source Inventory And Provenance

LumenCore records source identity, freshness, hash, row count, source class, and verification state. The system separates public data, buyer-authorized data, internal evidence, synthetic data, simulation, and unverified inputs so reviewers can see exactly what evidence supports a claim.

Baseline And Metric Lock

Before scoring a candidate workflow, LumenCore registers the incumbent baseline, accepted metric, allowed transformations, holdout window, exclusions, and success criteria. This reduces metric leakage, baseline switching, and retrospective cherry-picking.

Candidate Replay And Evaluation

Candidate workflows can be replayed against approved source bundles and compared with locked baselines. Negative results, uncertainty, missing data, and degraded performance are preserved rather than hidden.

Reviewer Evidence Manifest

Each run can produce a machine-readable manifest and a human-readable proof card showing source, baseline, candidate, metric, result, caveat, and next gate. This lets a technical reviewer inspect the evidence without relying on broad vendor language.

Controlled Pilot Transition

If a candidate survives replay, LumenCore converts the result into a bounded pilot plan: data access, owner, rollback path, reporting cadence, completion criteria, and claim boundaries.

Fit To NASA Data Center Infrastructure Modernization

NASA data center modernization decisions may involve high-density compute, hybrid/cloud evaluation, workload routing, resilience, cybersecurity posture, operational monitoring, energy/cost proxies, and AI-assisted operations. These decisions benefit from a validation layer that can compare options before deployment.

LumenCore can support:

- Source and telemetry inventory for infrastructure evidence.
- Baseline lock for current operations or incumbent infrastructure state.
- Candidate evaluation for AI operations, workload placement, cooling/energy heuristics, reliability workflows, and hybrid/cloud routing concepts.
- Evidence manifests that reviewers can inspect without trusting unbounded AI claims.
- Pilot-readiness gates for moving from concept to controlled test.

Suggested Limited Pilot Concept

NASA or an authorized partner selects one non-sensitive historical operations window, one incumbent baseline, and one candidate modernization workflow. LumenCore registers the evaluation plan, ingests approved source records, measures source quality, runs candidate-vs-baseline replay, and produces a reviewer packet.

Potential pilot outputs:

- Source inventory and quality ledger.
- Locked baseline and metric card.
- Candidate workflow replay summary.
- Resilience, operations, cost, or energy proxy scorecard where data permits.
- Negative-result and uncertainty register.

- Pilot recommendation: advance, revise, or stop.

Risk Controls

- No autonomous operational control without agency approval.
- No sensitive data in public proof artifacts.
- No training of commercial AI models on government data unless specifically authorized by the agency and allowed by contract.
- No guaranteed cost savings, energy savings, uptime gains, or cybersecurity outcomes.
- No claim of NASA validation, deployment, or endorsement.

- Human review before any external submission, pricing, or legal certification.

Why This Matters

Large infrastructure modernization decisions can fail when organizations cannot separate real measured evidence from demos, assumptions, retrospective benchmarks, or vendor claims. LumenCore gives reviewers a disciplined way to ask: what was measured, against what baseline, under what constraints, and with what reproducibility?

This is directly aligned with a market-research response because NASA can evaluate the approach without committing production access. The first value is a low-risk evidence protocol that can be scoped, inspected, and compared against other modernization concepts.

Attachments To Consider

- **FUNDING_REVIEWER_ZERO_FRICTION_PACK_2026-07-10.md**
- **TECHNICAL_GOV_REVIEWER_APPROVAL_STACK_2026-07-09.md**
- **MEASURED_SOURCE_EVIDENCE_REGISTER_2026-07-09.md**

- Public proof gateway: https://lumen-core.ai/proof_to_pilot.html

Final Human Gate

- Confirm official RFI instructions, recipients, subject line, deadline, page cap, and attachment rules from the live SAM notice.
- Robert approves the final response text, any company capability statements, and any past-performance language.
- Robert approves the final email send.